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EDS makeup session assignment - Theory.

- Write all ~~the~~ methods of pandas and its use.

→ ① Creating & Loading data.

- Key methods
 - Pd. Data Frame()
 - Pd. Series()
 - Pd. read_csv()
 - Pd. read_excel()
 - Pd. read_json()
 - Pd. read_sql()

- Real ~~life~~ life use

- Load student marks from excel
- Read bank transactions csv
- Fetch website data from database.

- Example:- `df = pd.read_csv("sales_data.csv")`

② Viewing & inspecting data

- Methods

- `df.head()`
- `df.tail()`
- `df.info()`
- `df.describe()`
- `df.shape`
- `df.columns`

- Real ~~life~~ life use:-

- ① Check first few rows of customer data
- ② Understand data types before ML

③ Data cleaning :-

- methods

- `df.drop()`
- `df.dropna()`
- `df.fillna()`
- `df.replace()`
- `df.rename()`

- real life use:-

- Remove null values in survey data
- Replace missing salary with average
- Rename confusing column names.

- Example:- `df.fillna(0, inplace=True)`

④ Selecting & filtering data

- Methods

- `df.loc[]`
- `df.iloc[]`
- `df[]` (column selection)
- `df.query()`

- Real-life Use

- Get students with marks > 80
- Extract specific rows for analysis
- Filter fraud transactions

- Example:- `df[df["marks"] > 80]`

⑤ Adding & modifying data

- Methods

- `df['New:col'] =`

- `df.assign()`

- `df.apply()`

- `df.map()`

- Real-life Use

- Add total marks column

- convert currency values

- Apply tax calculation

- Example:- `df["total"] = df["math"] + df["science"]`

⑥ Aggregation & statistics

- Methods

- `df.sum()`

- `df.mean()`

- `df.median()`

- `df.min(), df.max()`

- `df.count()`

- Real life Use

- calculate average salary

- Find total sales per month

- Analyze performance metrics

- ⑦ Grouping Data

- Methods

- `df.groupby()`
- `agg()`

• Real-Life Use

- Sales per city
- Students marks per class
- Revenue per product category

• Example:- `df.groupby("city")["sales"].sum()`

⑧ Merging & joining Data

• Methods

- `Pd.merge()`
- `df.join()`
- `Pd.concat()`

• Real-life Use

- combine customer + order data
- merge multiple Excel sheets
- join employee + salary tables.

⑨ Sorting & Ranking

• Methods

- `df.sort_values()`
- `df.sort_index()`
- `df.rank()`

• Real-life Use

- Rank top students
- Sort highest sales first

⑩ Time series handling.

• Methods.

- `pd.to_datetime()`
- `df.resample()`
- `df.shift()`

Real-life Use

- Stock market analysis
- Weather data tracking
- Daily app usage trends.

• ⑪ Real-life big use cases of Pandas.

① Banking

② E-commerce

③ Healthcare

④ Education

⑤ Data Science

(12) The most used types are:-

① read_csv

② head

③ Ioc

④ groupby

⑤ merge

⑥ apply

∴ These alone cover 80% of real-world work.